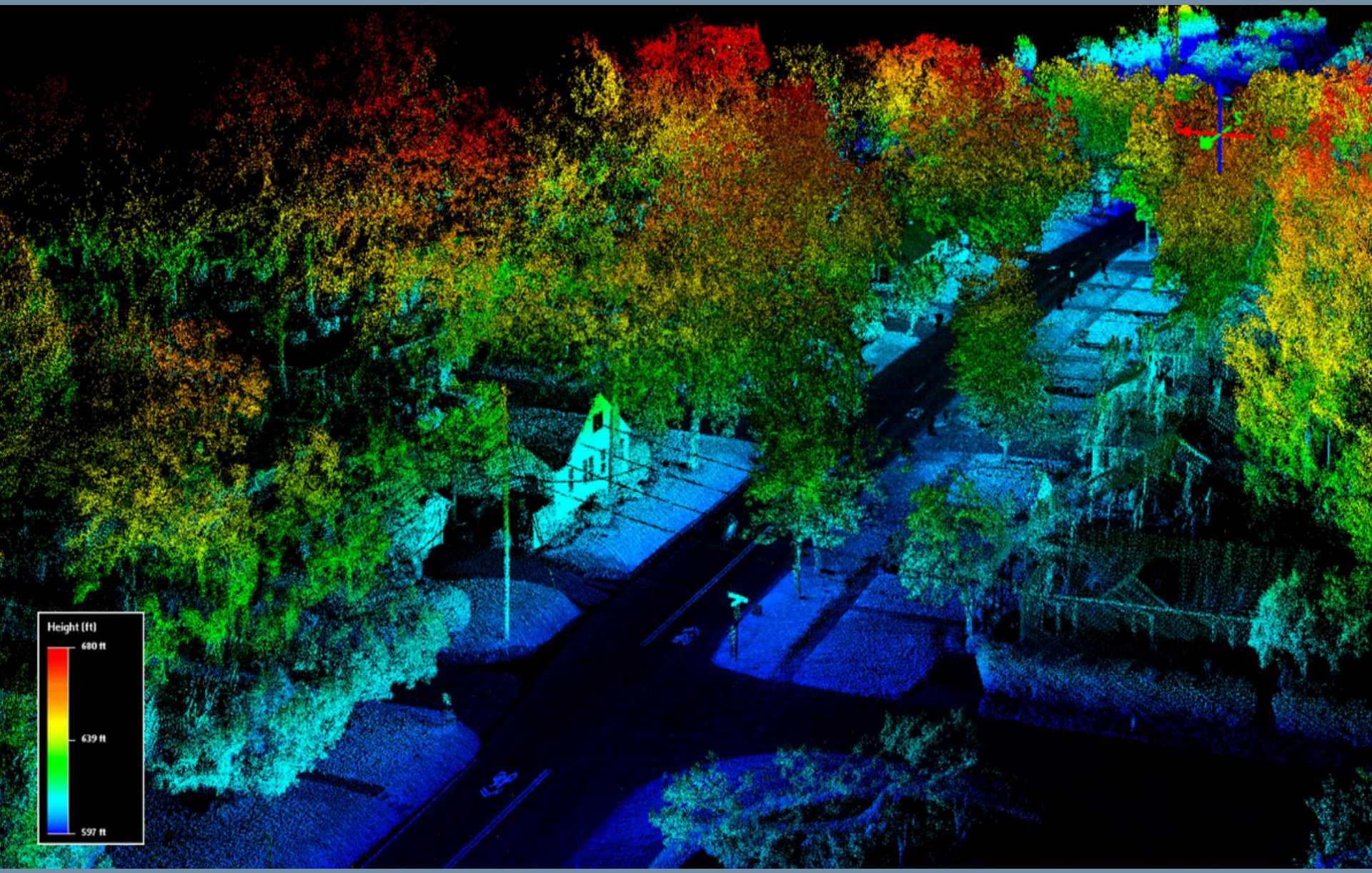
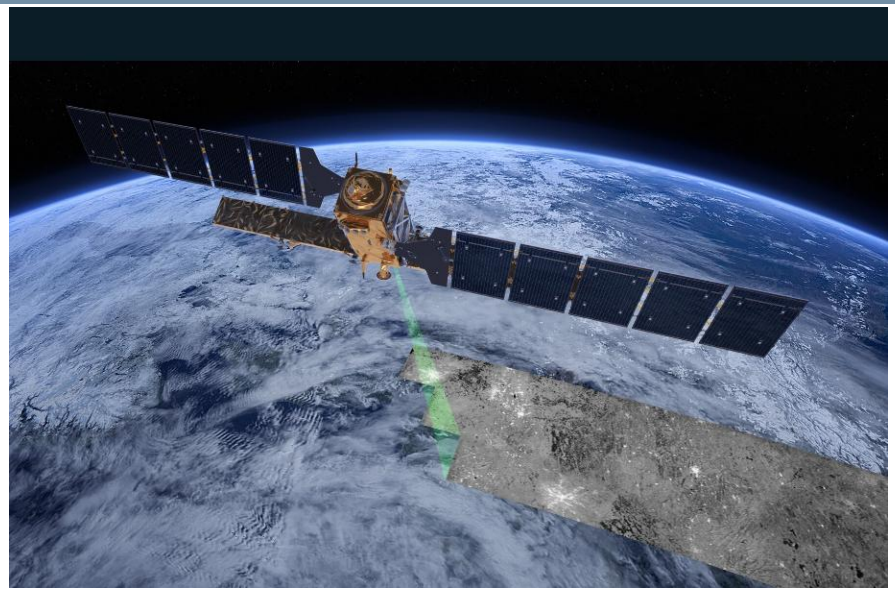
Coherent illumination, the antenna gain and resolution



Power and resolution: active remote sensing



This is Sentinel-1 an active imaging Radar (SAR) mission.

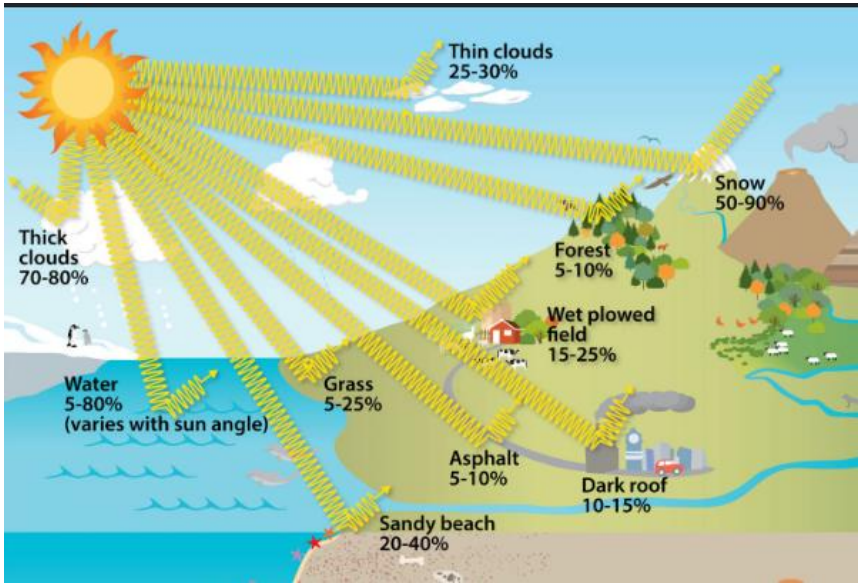
Assume:

- to transmit isotropically, radiating Pt = 1 kW,
- a resolution of 100 m^2,
- power reflection 10%
- carrier frequency is 5.6 GHz, bandwidth 50 MHz
- distance 1000 km
- surface temperature 290 K (17°)
- surface emissivity 90% (at microwaves)next

1) How much power 'illuminates' the resolution cell ?
2) How much power is scattered into the backward emisphere ?
3) Is this power significantly larger than the one emitted by the cell ?

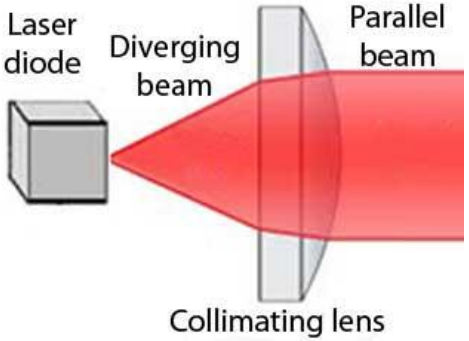
Resolution vs energy: wide or narrow beam ?

Black(or gray) body – **incoherent** emission

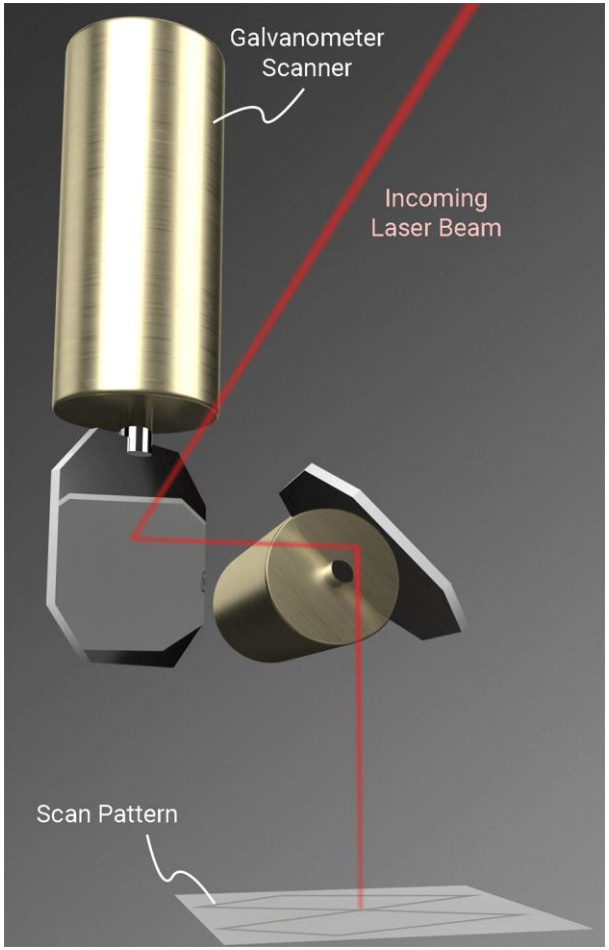


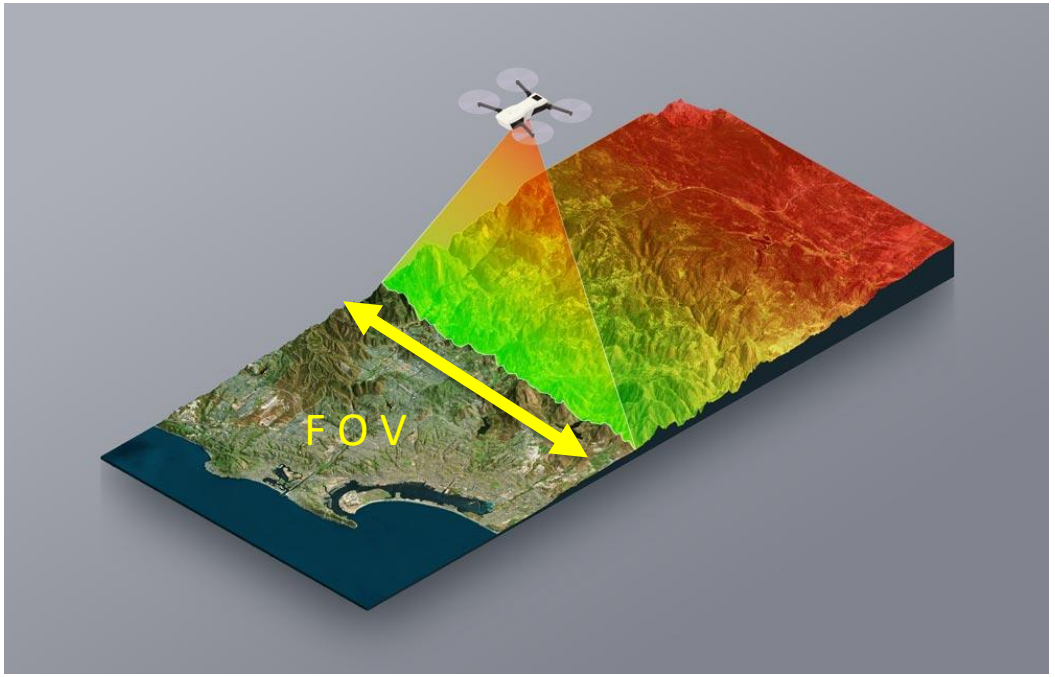
Laser – **Coherent** emission





Scanning the Field Of View: Light Detection And Ranging





A narrow, collimated beam and scanning over wide angles are essential for achieving simultaneously:

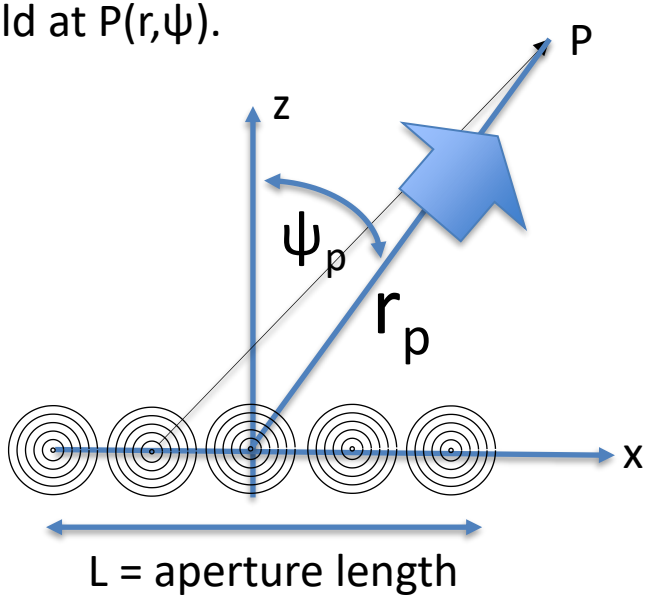
- resolution
- large FOV

Spatial coherence and antenna array

Let us assume a 2D geometry, and a set of point-like (isotropic) antenna along x-axis.

The goal is to generate a **strong, focused** field directed to P. How can we get this ?

Let first compute the field at P(r,ψ).

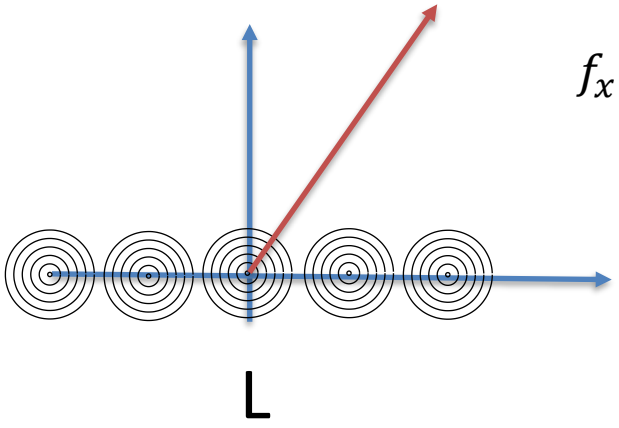


1) We evaluate the superposition of spherical waves (2D geometry shown for simplicity)

2) we assume all antennas transmitting the same frequency with fixed complex weights.

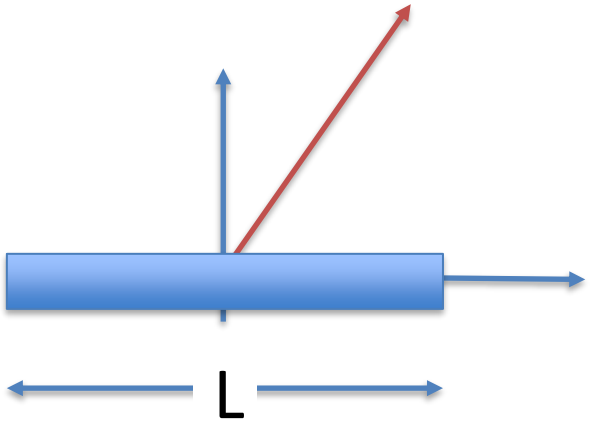
Comparison: array versus continuous (far field)

ARRAY

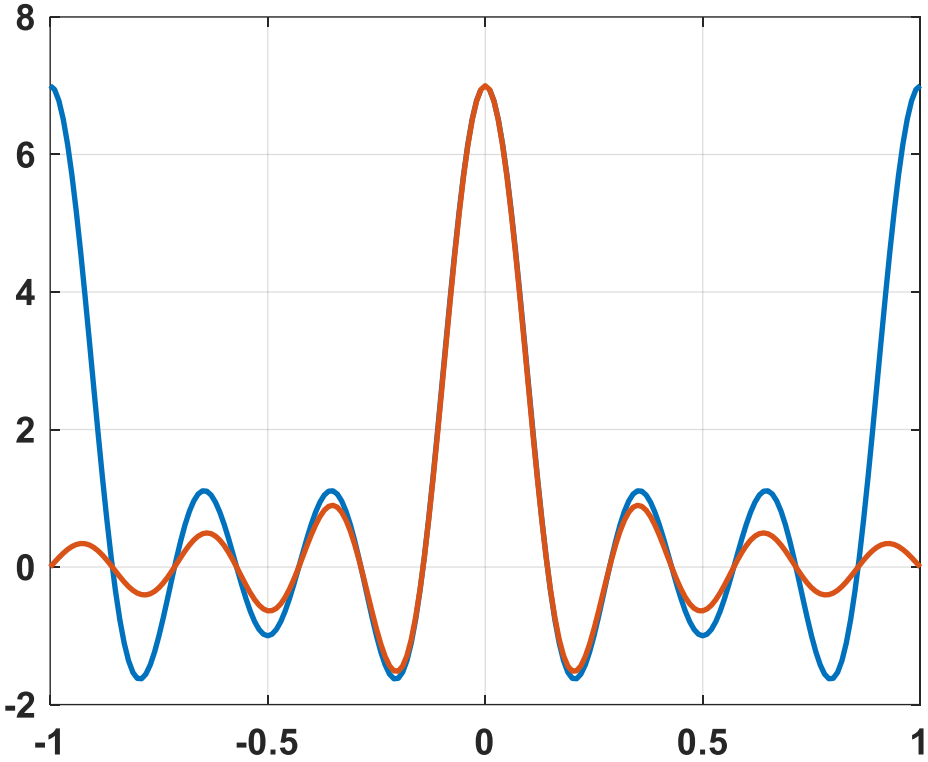


$$f_x = \frac{\sin\psi}{\lambda}$$

Continuous



Comparison: discrete illumination (array) versus continuous



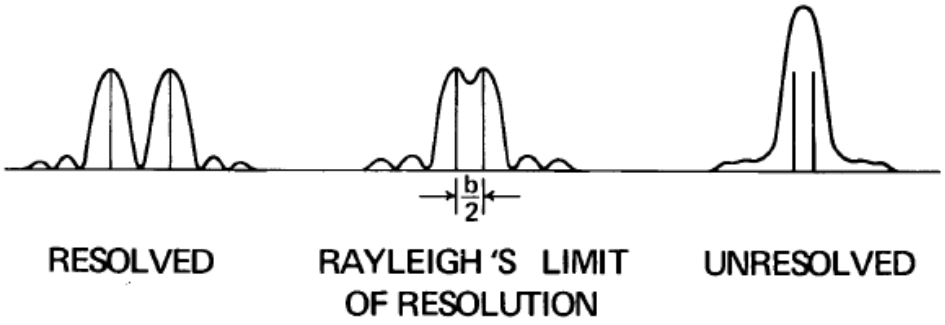
$$f_x = \frac{\sin\psi}{\lambda}$$

Can you tell me

1) the total length
2) the number of elements for the array
3) for which angles I have the first in the two cases
4) for which the grating lobes for the array

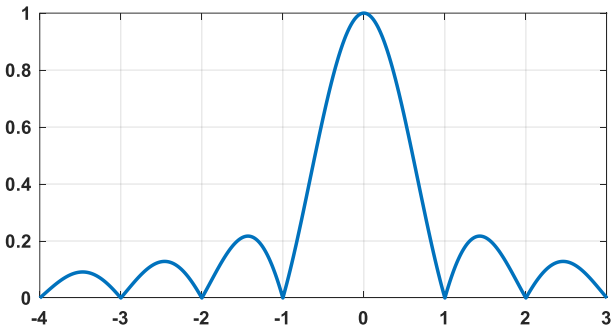
Resolution: Rayleigth

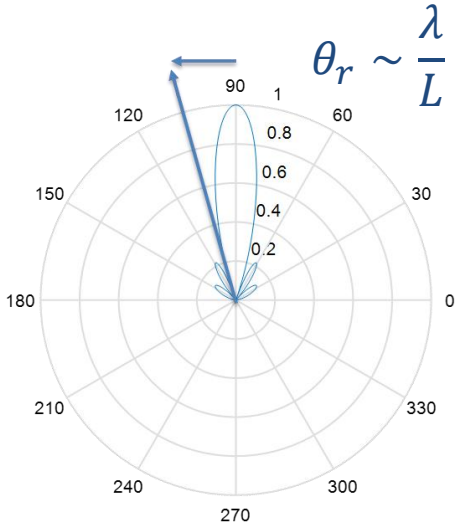
Geometric resolution is the minimum distance between two points sources such that their presence can be distinguished in the image.



Rayleigh's criterion: two point sources are resolved when the maximum of one pattern coincides with the first minimum of the other.
It was though for circular apertures.

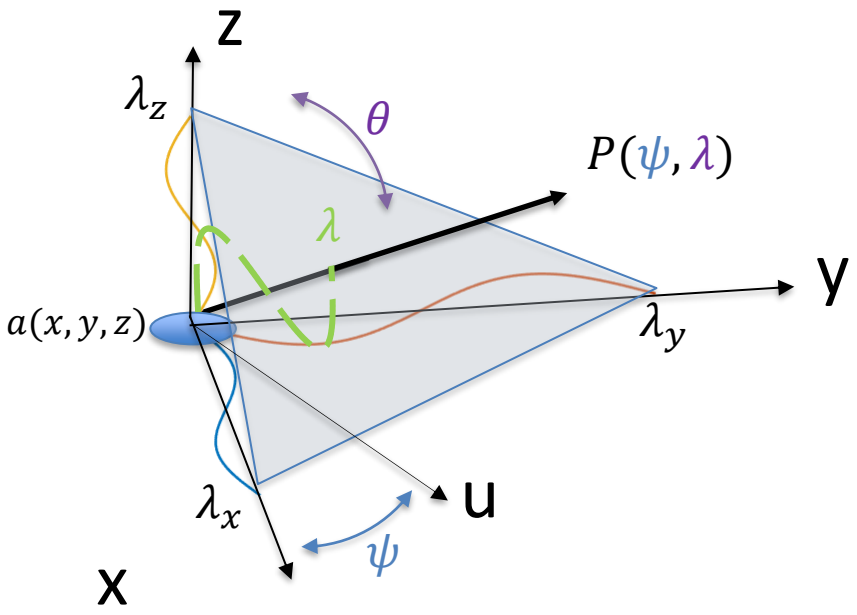
In our case ?





The 3Dimensional case

We assume that the aperture excitation at the antenna is a(x,y,z), and a monochromatic spherical wave. The wave can be approximated in the point P, direction ψ, λ by a plane wave directed like P.



Take note of angles in a standard spherical convention:

the polar angle, θ (also incidence angle)
the azimuth angle ψ

The amplitude of the wave in direction of P is proportional to the Fourier Transform:

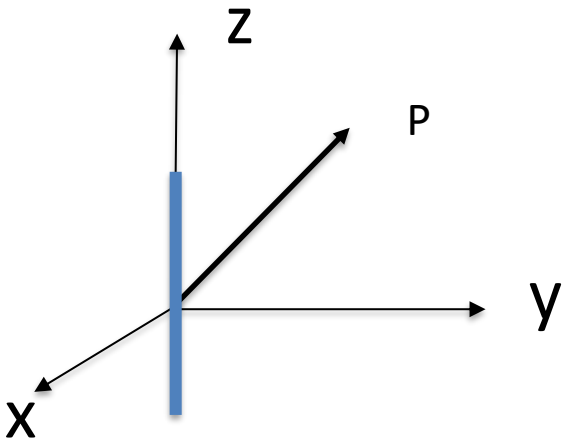
$$A(f_x, f_y, f_z)$$

where

$$f_x = \frac{1}{\lambda_x}; f_y = \frac{1}{\lambda_y}; f_z = \frac{1}{\lambda_z}$$

$$f = 1/\lambda = \sqrt{f_x^2 + f_y^2 + f_z^2}$$

Example: compute the gain of a dipole



Let us fix the direction $\mathrm{P}(\psi, \theta)$

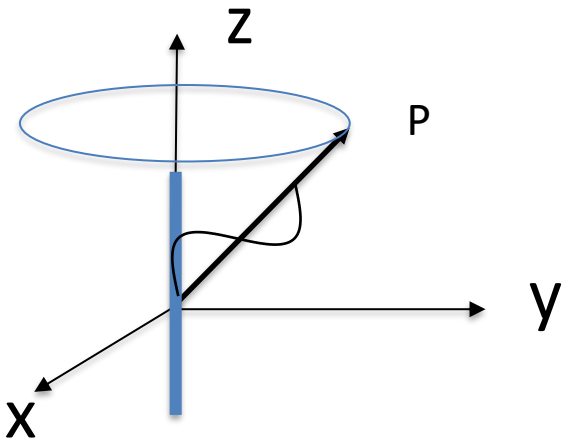
1) Compute the FT of the illumination

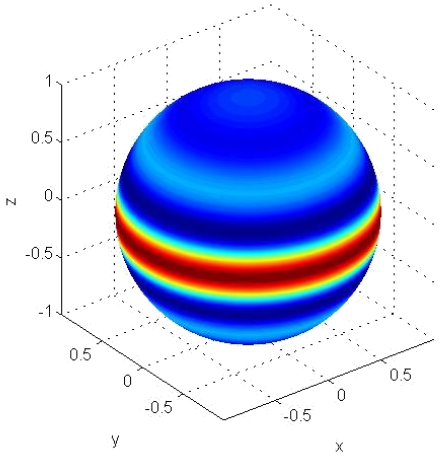
$$A(f_x, f_y, f_z)$$

2) Evaluate the FT for

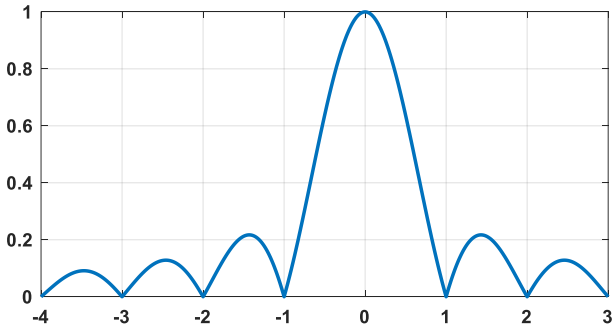
$$f_x = \frac{\sin\theta\cos\psi}{\lambda}; f_y = \frac{\sin\theta\sin\psi}{\lambda}; f_z = \frac{\cos\ \theta}{\lambda}$$

Example: compute the gain of a dipole



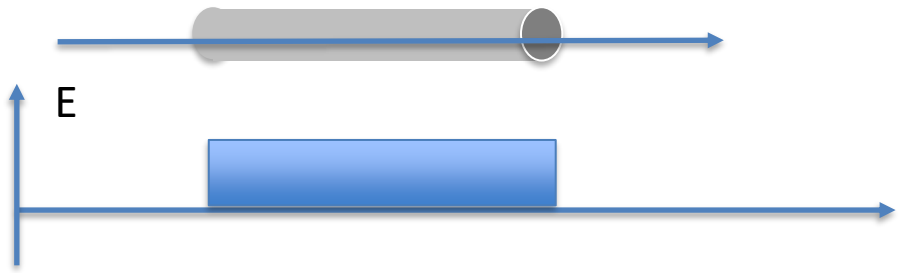


$$G(\theta) = L \cdot \left| sinc\left(L \frac{\cos\theta}{\lambda} \right) \right|;$$

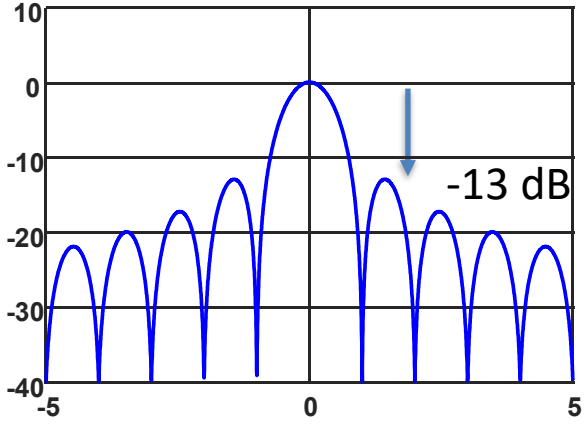


$$f_z = \frac{\sin\theta}{\lambda}$$

Triangular illumination



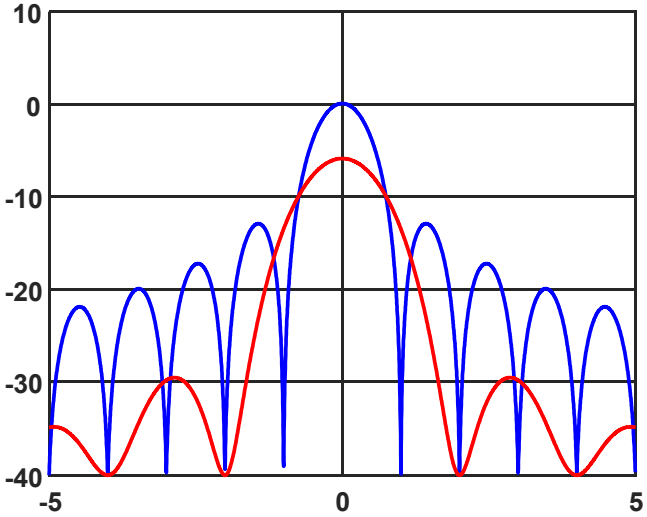
$$G(f_x) = sinc(L \cdot f_x)$$



The first sidelobe is at -13 dB.
If we want to improve it, we can apply triangular weighting.

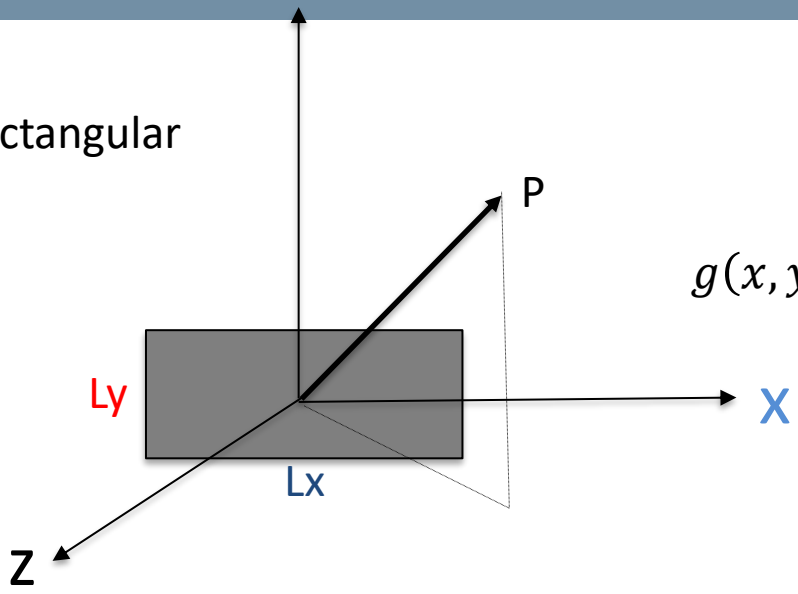


$$G_T(f_x) = sinc^2\left(\frac{L}{2} \cdot f_x\right)$$



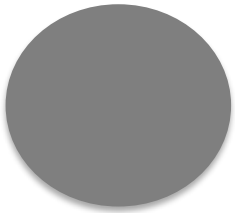
Examples

Uniform planar, rectangular



$$g(x,y) = rect\left(\frac{x}{L_x}\right) \cdot rect\left(\frac{y}{L_y}\right)$$

Uniform planar, circular ®



$$g(x,y) = ?$$